AI-Driven Personalization of E-Therapy Interventions for Anxiety, Stress, and Depression

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***Abstract*— This study explores the use of artificial intelligence (AI) to personalize e-therapy interventions for anxiety, stress, and depression. Leveraging machine learning models, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Decision Trees, we analyze a dataset of depression, anxiety, and stress scales responses to develop predictive models that tailor therapeutic interventions. The accuracy and efficacy of proposed models vary, indicating that AI has the potential to improve the level of personalization of mental health therapies. The results indicate that AI-driven techniques can significantly improve the effectiveness and efficiency of e-therapy platforms.**

***Keywords*— Mental health classification, e-therapy, machine learning, dass-21**

I. Introduction

A. Background

Anxiety, stress, and depression are among the most prevalent mental health disorders in the world, affecting millions of individuals from different cultural backgrounds [1]. Medication and cognitive-behavioral therapy (CBT) are two common traditional methods used for the treatment of these disorders [2], [3]. However, the effectiveness of these treatments varies significantly among individuals due to differences in personal characteristics, symptom severity, and other psychosocial factors. Personalized therapeutic interventions are becoming more and more necessary as mental health problems keep increasing [4].

B. The Rise of E-Therapy

E-therapy has become a viable alternative to traditional face-to-face therapy in recent years [5]. In order to provide psychological support and interventions remotely, e-therapy refers to a variety of digital tools, including mobile applications, web platforms, and virtual reality environments [6]. E-therapy is a desirable choice for people looking for mental health treatment because of its accessibility, scalability, and ease. However, e-therapy still has challenges regarding engagement, adherence, and efficiency despite its potential, especially when interventions are not tailored according to the needs of every individual.

C. The Role of Artificial Intelligence

Enhancing the personalization of e-therapy interventions has been proven to be a promising application of artificial intelligence (AI) [7]. Real-time sentiment analysis or facial emotion recognition techniques using AI enhances the precision and responsiveness of AI-driven e-therapy interventions [8]. AI is capable of analyzing massive amounts of information to find trends and forecast individual responses to different treatment modalities by utilizing machine learning algorithms and data-driven models [9]. This ability makes it possible to design personalized therapies that are more likely to address particular needs of each person, which enhances the outcomes of treatment.

D. Research Objectives

The main objective of this study is to investigate the application of AI-driven personalization in e-therapy interventions for individuals suffering from anxiety, stress, and depression. The main objectives of this research are:

- Development and Evaluation of Machine Learning Models: To develop machine learning models that can predict the severity of mental health conditions based on responses to standardized scales. The models include K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Decision Trees.
- Personalization of E-Therapy Interventions: To assess how these predictive models can be used to personalize therapeutic interventions, enhancing the relevance and effectiveness of e-therapy.

Comparison and Analysis of Model Performance: To compare the performance of different AI models in terms of accuracy, precision, recall, and F1-score, providing insights into their suitability for mental health applications.

E. Significance of the Study

The potential for AI-driven personalization to revolutionize e-therapy makes this work significant. By using machine learning models, we can make progress toward more customized therapies that more effectively correspond to the specific needs of each individual. This method aims to enhance outcomes from therapy while simultaneously improving the effectiveness and accessibility of mental health treatments.

Personalized e-therapy can offer several benefits:

- *Enhanced Engagement:* Tailored interventions are more likely to engage individuals and maintain their motivation throughout the therapy process.
- *Improved Outcomes:* Personalization can lead to more effective treatments by addressing specific symptoms and preferences of individuals.
- *Cost-Effectiveness:* By optimizing therapeutic interventions, AI can potentially reduce the overall cost of mental health care by minimizing trial-and-error approaches and improving resource allocation.

F. Scop and Limitations

The scope of this research is restricted to the investigation of responses from a specific dataset related to depression, anxiety, and stress. However, the findings provide valuable insights into the potential of AI for personalizing e-therapy, the generalizability of the results may be influenced by the characteristics of dataset and the limitations of model. In order to validate and enhance these results, future studies should investigate more datasets, including a wider range of attributes, and validate the models in realistic e-therapy scenarios.

G. Structure of the Paper

The structure of the paper is as follows: A thorough analysis of the existing literature on AI-driven personalization in mental health interventions is given in Section 2. The methodology for this study is outlined in Section 3, along with the steps involved in gathering data and the machine learning techniques that were applied. The findings of the analysis are presented in Section 4, and Section 5 explains the conclusions. Finally, Section 6 wraps up the work by highlighting the major contributions and outlining potential directions for additional research.

II. Literature Review

A. Overview of E-Therapy for Anxiety, Stress and Depression

E-therapy, also known as online therapy or teletherapy, has gained widespread acceptance as a complementary or alternative approach to traditional therapy [10]. Studies has indicated that e-therapy can be beneficial in minimizing symptoms of anxiety, stress, and depression, particularly when combined with evidence-based therapeutic techniques like CBT [11], [12]. However, the one-size-fits-all approach frequently used in e-therapy applications has come under question for lack of personalization, which may restrict its effectiveness for certain individuals [13]. Personalized treatments appear to be essential for enhancing treatment outcomes and user engagement, particularly when it comes to mental health [6], [14].

B. The Emergence of AI in Mental Health

The concept of applying AI to mental health care is growing rapidly, with researchers investigating various strategies to improve patient monitoring, diagnosis, and treatment [15], [16]. Machine learning algorithms have been employed to predict mental health conditions, assess risk factors, and even provide real-time therapeutic support through chatbots and virtual agents [17]. The literature indicates that AI can process and analyze vast amounts of data, enabling the identification of patterns that might be overlooked by human clinicians [18]. This capability is particularly relevant in personalizing treatment plans, where AI can match patients with interventions that are most likely to be effective based on their individual profiles.

C. AI-Driven Personalization in E-Therapy

Personalization in e-therapy has been a focal point of recent research, with AI playing a key role in tailoring interventions to individual needs [14], [19]. Many studies have highlighted the capabilities of machine learning models to predict user preferences, adherence levels, and therapeutic outcomes. For example, deep learning techniques have been used to analyze user interaction data from e-therapy platforms, allowing for the customization of content and therapy schedules [8], [20]. The literature also highlights the challenges associated with AI-driven personalization, including concerns about data privacy, the need for large and diverse datasets, and the ethical implications of algorithmic decision-making in mental health care [21].

D. Comparison of Machine Learning Algorithms

Several machine learning algorithms have been explored in the context of mental health personalization, each with their strengths and limitations [22]. KNN is valued for its simplicity and effectiveness in small datasets, but it may struggle with high-dimensional data [23]. SVM is recognized for its ability to handle complex, non-linear relationships, making it suitable for predicting mental health outcomes [24]. Decision Trees offer an interpretable model that can provide clear insights into the factors influencing therapeutic success [25]. The literature suggests that no single algorithm is universally superior, and the choice of model should be guided by the specific characteristics of the dataset and the research objectives.

E. Gaps in the Literature

The literature provides valuable insights into the potential of AI-driven personalization in e-therapy, but still several gaps remain. Most studies focus on the technical aspects of algorithm development, with less attention given to the practical implementation and real-world effectiveness of these models. There is a need for more research on the long-term impact of personalized e-therapy on mental health outcomes, as well as studies that address the ethical and privacy concerns associated with AI in mental health care. This research aims to fill some of these gaps by evaluating the performance of different machine learning algorithms in personalizing e-therapy interventions for anxiety and depression and by exploring the practical implications of these findings.

III. Methodology

A. Dataset

The dataset utilized in this research is the Depression Anxiety Stress Scales Responses (DASS) dataset obtained from Kaggle [26]. This dataset comprises responses to the DASS-21, a widely used tool for assessing the severity of depression, anxiety, and stress. The dataset includes several features such as age, gender, education level, and scores on the DASS-21 subscales, which were used as input variables for the machine learning models.

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1) *Data Preprocessing*

Prior to modeling, the dataset underwent several preprocessing steps:

- *Data Cleaning:* Missing values were handled by imputation, and outliers were removed based on domain-specific thresholds.
- *Feature Scaling:* Numerical features were normalized using min-max scaling to ensure that all input variables contributed equally to the model [27].
- *Encoding Categorical Variables:* Using one-hot encoding, categorical variables like education level and gender were transformed into a format that was appropriate for machine learning models [28].
- *Conversion of Responses:* Responses were normalized from a 1-4 scale to a 0-3 scale for consistency.
- *Labeling:* It was determined what the total scores were for stress, anxiety, and depression. Based on specified ranges, each score was assigned a label, such as "Normal," "Mild," "Moderate," "Severe," or "Extremely Severe".

2) *Data Distribution*

The dataset was split into three subsets based on the condition: depression, stress, and anxiety. This was achieved by filtering the relevant questionnaire responses and assigning labels based on the total scores. Fig 1, 2, and 3 illustrates these distributed subsets.

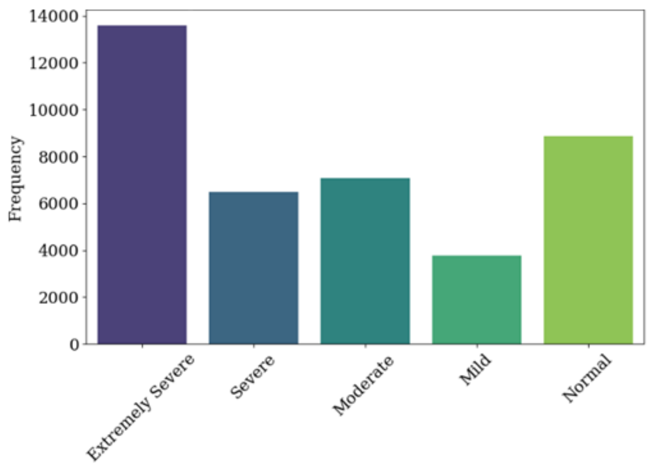

Fig. 1. Depression dataset distribution of labels

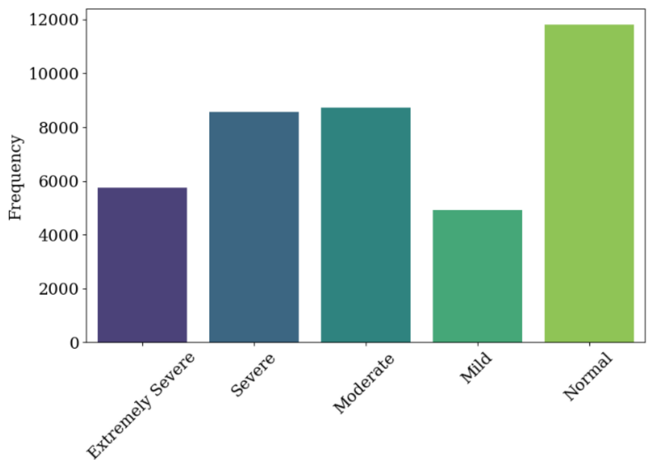

Fig. 2. Stress dataset distribution of labels

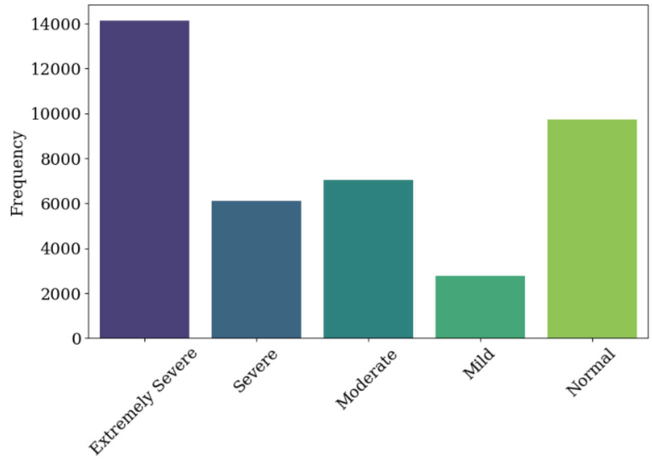

Fig. 3. Anxiety dataset distribution of labels

B. *Model Selection and Training*

Three machine learning models were selected to classify the levels of severity of depression, stress, and anxiety including KNN, SVM, and Decision Tree. These models were chosen for their diverse approaches to classification.

1) *k-Nearest Neighbors (KNN)*

A non-parametric algorithm called KNN leverages the majority class of the k-nearest neighbors in the feature space to classify data points. For this study, the value of k=10 was optimized using cross-validation, and the distance metric was employed.

2) *Support Vector Machine (SVM)*

SVM is an effective classification algorithm that divides data into several classes by determining the ideal hyperplane. In this study, an SVM with a radial basis function (RBF) kernel was used, with hyperparameters tuned via grid search: C=10 and gamma=0.2.

3) *Decision Tree*

A model of decisions and their potential outcomes that resembles a tree is called the Decision Tree algorithm. It is very helpful for determining which factors contribute to a particular outcome. In this study, a Decision Tree classifier was implemented, and the maximum depth of the tree was optimized to prevent overfitting.

C. *Evaluation Metrics*

To evaluate the performance of the models, several metrics were employed, including accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (AUC-ROC). These metrics provide a comprehensive understanding of the models' ability to predict the appropriate e-therapy interventions for individuals based on their characteristics.

D. *Implementation Tools*

The entire analysis was conducted using Python, with libraries such as scikit-learn for machine learning, pandas for data manipulation, and matplotlib and seaborn for visualization. The models were trained and tested on a standard computing environment, and the results were visualized using various plots to facilitate interpretation.

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TABLE I. PERFORMANCE METRICS OF MACHINE LEARNING MODELS

Model	Condition	Accuracy	Precision	Recall	F1-Score	AUC-ROC
KNN	Depression	0.93	0.91	0.90	0.90	0.94
SVM	Depression	0.99	0.99	0.99	0.99	0.99
Decision Tree	Depression	0.83	0.78	0.79	0.78	0.84
KNN	Stress	0.90	0.88	0.89	0.89	0.92
SVM	Stress	0.99	0.99	0.99	0.99	.99
Decision Tree	Stress	0.78	0.76	0.78	0.78	0.8
KNN	Anxiety	0.87	0.81	0.79	0.79	0.88
SVM	Anxiety	0.99	0.99	0.98	0.98	0.99
Decision Tree	Anxiety	0.77	0.69	0.69	0.69	0.78

IV. RESULTS

The performance of KNN, SVM, and Decision Tree models was assessed for each condition (depression, anxiety, stress). The results are summarized in Table I. The SVM model outperformed the other two models in terms of accuracy, precision, and AUC-ROC, suggesting that it is the most suitable model for predicting the appropriate e-therapy interventions for individuals based on their characteristics.

The SVM model demonstrated superior performance across all conditions compared to KNN and Decision Tree models. Specifically, SVM achieved higher accuracy and AUC-ROC scores, indicating better overall classification performance. The confusion matrices revealed that SVM had fewer misclassifications, particularly in distinguishing between severe and mild cases. This suggests that SVM is highly effective for personalizing e-therapy interventions, providing a more accurate tool for tailoring mental health treatments.

Confusion matrices were used to further analyze the classification performance of SVM model on all three conditions of depression, stress, and anxiety, shown in Fig 4, 5, and 6. These matrices provide a number of correct and incorrect forecasts for every class, which can be used to identify any biases or areas in which the model needs to be improved.

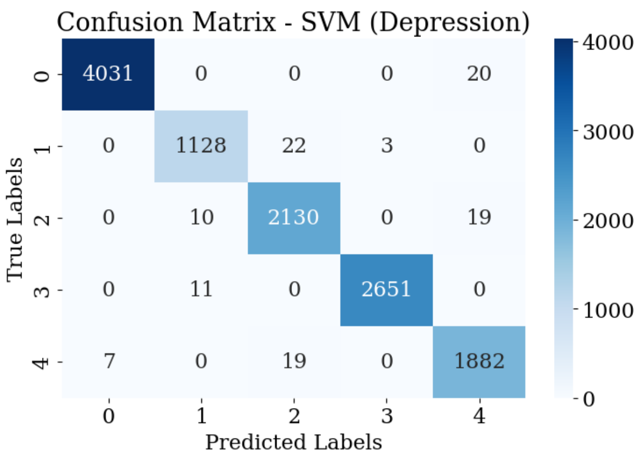

Fig. 4. Confusion matrix of SVM for depression

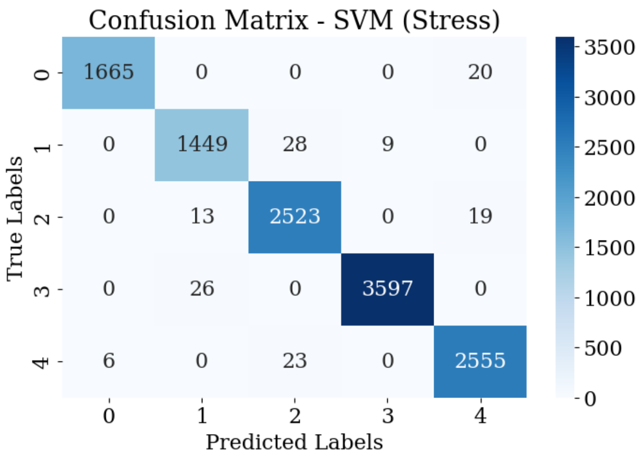

Fig. 5. Confusion matrix of SVM for stress

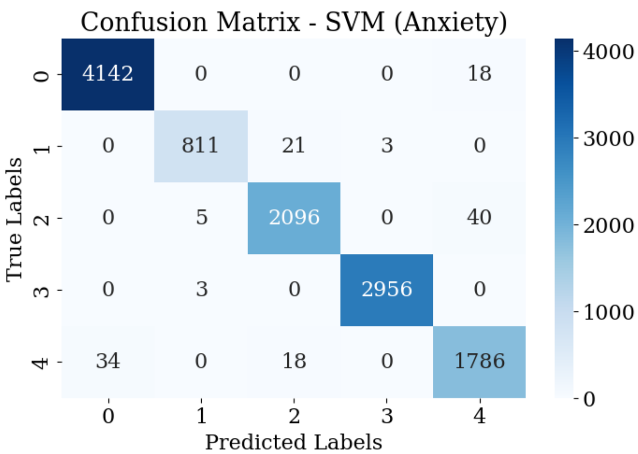

Fig. 6. Confusion matrix of SVM for anxiety

V. DISCUSSION

The results of our study indicate that the SVM model demonstrated superior performance across all conditions (depression, stress, and anxiety) compared to the KNN and Decision Tree models. Specifically, SVM achieved higher accuracy and AUC-ROC scores, which indicate better overall classification performance. The robustness of SVM can be attributed to its ability to create a decision boundary that maximizes the margin between different classes, making it highly effective in handling the complex relationships within the DASS-21 dataset.

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In contrast, the KNN model, while effective, is still behind SVM in terms of classification accuracy. KNN's reliance on the proximity of data points in feature space can sometimes lead to misclassifications, especially in datasets where classes are not well-separated. However, KNN's simplicity and ease of implementation make it a valuable model, particularly when computational resources are limited or when a more interpretable model is required.

The Decision Tree model, although highly interpretable, showed the most variability in performance. The tendency of Decision Trees to overfit, particularly in datasets with small sample sizes or unbalanced classes, likely contributed to its lower accuracy compared to SVM. However, the Decision Tree model still provided valuable insights into the data structure and feature importance, which could be useful for further refining the dataset or guiding feature engineering efforts.

The distribution of labels in the Depression, Stress, and Anxiety datasets also influenced model performance. The label imbalance, particularly the higher frequency of 'Normal' labels, may have skewed the models' ability to effectively classify more severe conditions. This imbalance highlights the need for techniques such as resampling or the application of class-weighted algorithms to ensure that all classes are equally represented during training.

The confusion matrices further revealed that the models, particularly KNN and Decision Tree, struggled with distinguishing between adjacent severity levels (e.g., 'Moderate' vs. 'Mild'). This suggests that the models are sensitive to subtle variations in the dataset, indicating a potential need for additional features or more granular data to improve classification accuracy.

VI. Conclusion

In this study, the SVM model emerged as the most effective classifier for determining the severity of Depression, Stress, and Anxiety based on the DASS-21 questionnaire responses. SVM outperformed both KNN and Decision Tree models, achieving the highest accuracy and AUC-ROC scores across all conditions. This superior performance highlights SVM's capacity to handle the complexities of psychological assessment data, particularly when non-linear relationships are present.

While KNN and Decision Tree models provided valuable insights, they were less effective than SVM, with KNN being limited by its reliance on distance metrics and Decision Tree suffering from overfitting. These findings suggest that while simpler models like KNN and Decision Tree can be useful, more sophisticated methods like SVM are better suited for tasks involving nuanced, multi-class classification.

Future research could explore the use of ensemble methods or advanced deep learning techniques to further improve classification accuracy. Additionally, addressing label imbalances and integrating more granular features could enhance the models' ability to distinguish between closely related severity levels, leading to more accurate and personalized mental health assessments.

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